MAE 3240 Heat Transfer

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Name: Charlie Pearson

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Problem set number: #2

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Acknowledgement:

No help

Question 1

1) Known:

- $R = 0.5\,m$
- $H = 1\,m$
- $T_s = 50°C$ (exterior wall)
- $T_\infty = 20°C$
- $h = 15\,W/m^2K$
- $L = 0.02\,m$ (thickness of insulation)
- $k = 0.1\,W/mK$ (k of insulation)

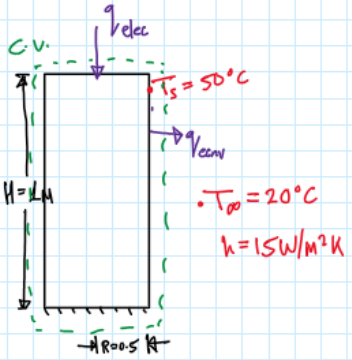
} Part b

Find:

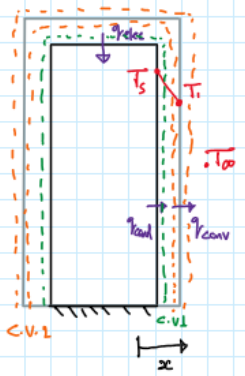
a & b: q to maintain constant water temp 50°C

Schematic:

a)



b)



$q_{cond} = q_{conv}$ (c.v. across insulation)

$q_{elec} = q_{cond}$ (c.v. across outer wall)

Properties:

- $h = 15\,W/m^2K$
- $k = 0.1\,W/mK$

Assumptions:

- Steady state
- Wall of heater thin + conductive → $T_s = T_{water} = 50°C$
- Bottom of heater insulated
- Thermal radiation negligable
- Temp distribution in insulation foam is linear ∴ $\frac{dT}{dx} \simeq \frac{T_1 - T_s}{L}$
- 1D conduction across insulating foam

Analysis:

a) Steady state assumption yields the following:

$$\dot{E}_{st} = \dot{E}_{in} + \dot{E}_{gen} - \dot{E}_{out} = 0$$

$$\dot{E}_{in} = 0$$

$$\dot{E}_{gen} = q_{elec}$$

$$\dot{E}_{out} = q_{conv}$$

$$\Rightarrow q_{elec} = q_{conv}$$

Solving for q_{elec}:

$$q_{elec} = Ah(T_s - T_\infty)$$

$$= (2\pi RH + \pi R^2)h \cdot (T_s - T_\infty)$$

$$= (2\pi \times 0.5 \times 1 + \pi \times 0.5^2) \times 15 \times (50 - 20)$$

$$= 1767 \text{ W}$$

$$\boxed{q_{elec} = 1.77 \text{ kW}}$$

b) From surface energy balance at insulation-air interface:

$$q_{cond} = q_{conv}$$

$$\Rightarrow -kA\frac{dT}{dx} = hA(T_1 - T_\infty)$$

$$-k \cdot \frac{T_1 - T_s}{L} = h(T_1 - T_\infty)$$

Rearrange for T_1:

$$T_1 - T_s = -\frac{hL}{k}(T_1 - T_\infty)$$

$$T_1 + \frac{hL}{k}T_1 = \frac{hL}{k}T_\infty + T_s$$

$$T_1 = \frac{\frac{hL}{k}T_\infty + T_s}{1 + \frac{hL}{k}}$$

$$T_1 = \frac{\frac{15 \times 0.02}{0.1} \times 20 + 50}{1 + \frac{15 \times 0.02}{0.1}}$$

$$T_1 = 27.5°C$$

Now doing energy balance for water tank (similar to in a)

$$q_{elec} = q_{cond}$$

$$q_{elec} = -kA\frac{dT}{dt}$$

$$= -kA\frac{T_1 - T_s}{L}$$

$$= -k(2\pi RH + \pi R^2) \cdot \frac{T_1 - T_s}{L}$$

$$= -0.1(2 \times \pi \times 0.5 \times 1 + \pi \times 0.5^2) \cdot \frac{27.5 - 50}{0.02}$$

$$= 442W$$

$$\boxed{q_{elec} = 442W}$$

c) The power required from the heater dropped by around 75% when the insulation was installed. This shows how important insulation is when it comes to the efficiency of water heaters.

Question 2

2) Known:

- $L = 0.1\,M$
- $k = 1\,W/(M \cdot K)$
- $T_\infty = 20°C$
- $h = 100\,W/(M^2 \cdot K)$
- $q'' = 1000\,W/M^2$

Find:

- Temp of inner surface $(x=0)$, T_1
- Temp of exterior surface $(x=L)$, T_2

} a

- Max heat flux of heater without overheating } b

Shematic:

Properties:

- $k = 1\ W/(m \cdot K)$
- $h = 100\ W/(m^2 \cdot K)$

Assumptions:

- System is in S.S.
- 1D heat transfer
- Temp distribution inside plastic layer
- Heater infinitely thin
- Radiation is neglegable

Analysis:

Under S.S. conditions, the energy balance across a c.v. is given by:

$\dot{E}_{in} + \dot{E}_{gen} - \dot{E}_{out} = 0$

Applying this to the heater & plastic layer where $\dot{E}_{gen} = q_{elec}$ and $\dot{E}_{out} = q_{cond}$

$\Rightarrow q''_{elec} = q''_{cond}$

Also applying this to the c.v. at the plastic-coolant boundary gives $\dot{E}_{in} = q_{cond}$, $\dot{E}_{out} = q_{conv}$

$\Rightarrow q''_{cond} = q''_{conv}$

This therefore yields $q''_{elec} = q''_{cond} = q''_{conv}$

Solve $q''_{conv} = q''_{elec}$ for T_2 :

$$h(T_2 - T_\infty) = q''_{elec}$$

$$T_2 = \frac{q''_{elec}}{h} + T_\infty$$

$$T_2 = \frac{1000}{100} + 20$$

$$\boxed{T_2 = 30\,°C}$$

Now solving $q''_{cond} = q''_{elec}$ for T_1

$$-k\frac{dT}{dx} = q''_{elec}$$

$$-k\frac{T_2 - T_1}{L} = q''_{elec}$$

$$T_2 - T_1 = -\frac{q''_{elec}L}{k}$$

$$T_1 = T_2 + \frac{q''_{elec}L}{k}$$

$$T_1 = 30 + \frac{1000 \times 0.1}{1}$$

$$\boxed{T_1 = 130\,°C}$$

b) similarly to part (a), $q''_{elec} = q''_{cond} = q''_{conv}$, where $T_{1,max} = 200\,°C$

Solve $q''_{cond} = q''_{conv}$ for T_2

$$-k\frac{T_2 - T_{1,max}}{L} = h(T_2 - T_\infty)$$

$$T_2 - T_{1,max} = -\frac{hL}{k}(T_2 - T_\infty)$$

$$T_2 + \frac{hL}{k}T_2 = \frac{hL}{k}T_\infty + T_{1,max}$$

$$T_2 = \frac{\frac{hL}{k}T_\infty + T_{1,max}}{1 + \frac{hL}{k}}$$

$$T_2 = \frac{\frac{100 \times 0.1}{1} \times 20 + 200}{1 + \frac{100 \times 0.1}{1}}$$

$$T_2 = 36.4\,°C$$

Now solving $q''_{conv} = q''_{elec}$

$$h(T_2 - T_\infty) = q''_{elec}$$

$$100(36.4 - 20) = q''_{elec}$$

$$q''_{elec} = 1636\ W/m^2$$

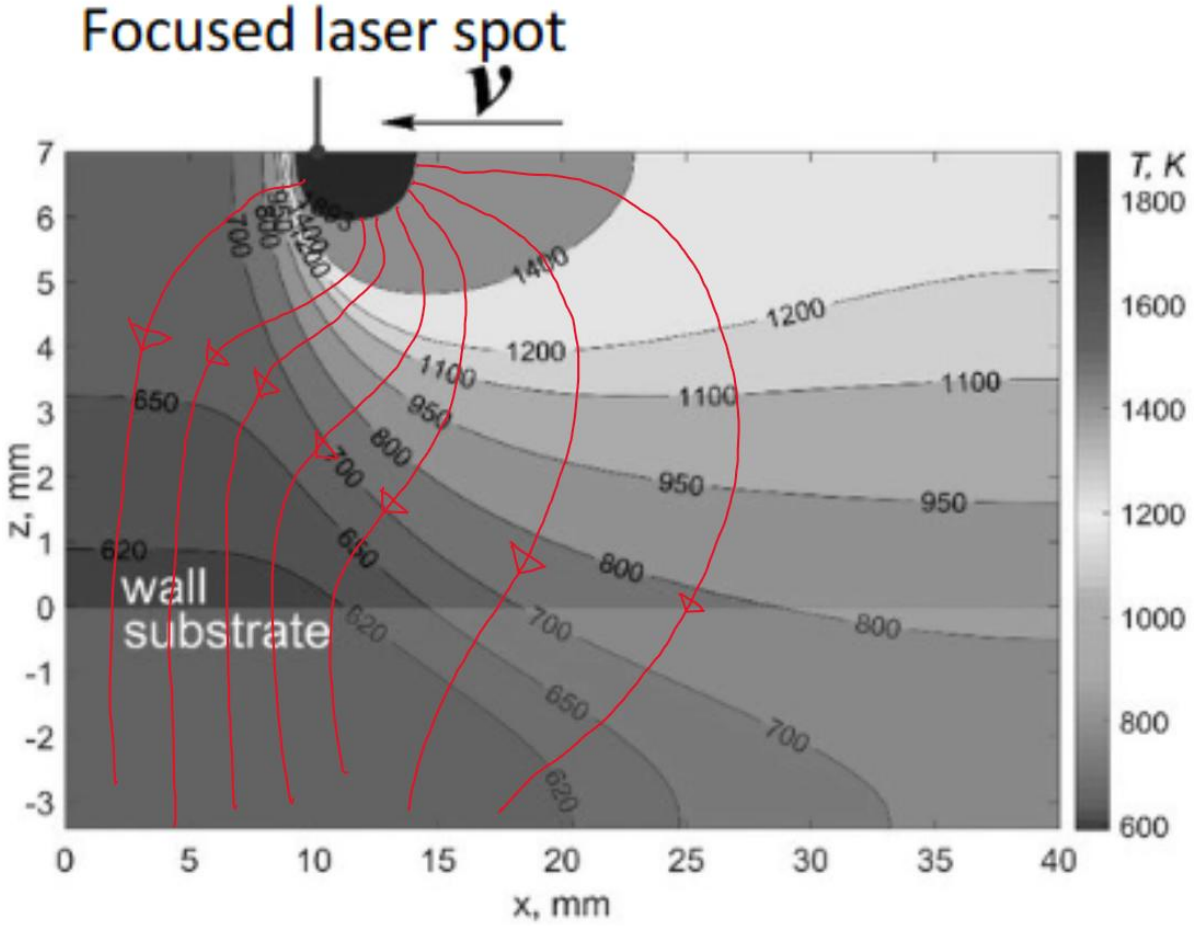
c) Want to increase the rate at which flux travels through the system.

Method 1: Increase the heat transfer coefficient, h in the coolant (e.g. using forced convection of a liquid) → increasing convection rate.

Method 2: Change the Plastic to a more conductive material e.g. metal. This will increase the rate of conduction.

Method 3: Decrease the temperature of the coolant, T_∞. This will increase the convection rate by increasing the temperature differential.

Question 3



(Heat flux vectors travel perpendicular to isotherms, down the temperature gradient.)

b) $\vec{q}'' = q''_n \hat{n} = -k \frac{dT}{dn} \hat{n}$

where $\hat{n}$ is the unit vector in the direction of heat transfer (perpendicular to the isotherms).

If you draw a vector $\vec{q}''_n \hat{n}$ perpendicular to the isotherms on the left & right side of the spot, the rate of heat change along the vector is larger on the left side than the right $\left(\frac{dT}{dn}\Big|_{Left} > \frac{dT}{dn}\Big|_{right} \right)$. This can be seen by how close together the isotherms on the left are compared to the right. Therefore the heat flux is larger on the left of the spot than the right.